

Matrix Multiplication

Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Determine whether two matrices can be multiplied by checking inner dimensions
- Identify the order (dimensions) of the resulting product matrix
- Perform matrix multiplication by multiplying rows of the first matrix by columns of the second matrix

Problems

1. Can these two matrices be multiplied in the order $A \times B$? Write Yes or No, and state the order of the product if possible. Matrix A is 2×3 and Matrix B is 3×4 .

$$(2 \times 3) \cdot (3 \times 4)$$

2. Can these two matrices be multiplied in the order $A \times B$? Write Yes or No, and state the order of the product if possible. Matrix A is 3×2 and Matrix B is 3×2 .

$$(3 \times 2) \cdot (3 \times 2)$$

3. Find the product of the two matrices shown below.

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

4. Compute the matrix product $A \times B$ where A and B are given below.

5. Compute the matrix product $A \times B$ where A is a 2×3 matrix and B is a 3×2 matrix, as shown below.

6. Compute the matrix product $A \times B$ where A is 2×2 and B is 2×3 , as shown below. State the order of the resulting matrix.



7. Determine whether $B \times A$ is defined for the matrices below. If it is, compute the product.

8. Compute A^2 (A times A) for the matrix A shown below.

9. Given the three matrices below, compute $(A \times B) \times C$. First find $A \times B$, then multiply the result by C.

10. Given matrices A and B below, compute $A \times B$ and then verify that in general $A \times B \neq B \times A$ by computing $B \times A$ as well.



Matrix Multiplication — Answer Key

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Answer Key

1. Answer: Yes; product is 2×4

- Check inner dimensions: the number of columns of A (3) equals the number of rows of B (3). ✓
- The product matrix order is given by the outer dimensions: 2×4 .

2. Answer: No; inner dimensions do not match ($2 \neq 3$)

- Check inner dimensions: columns of A = 2, rows of B = 3.
- Since $2 \neq 3$, the multiplication is not defined.

3. Answer: $[[2,3],[1,4]]$

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

- Multiply row 1 of A by each column of I: $[2 \cdot 1 + 3 \cdot 0, 2 \cdot 0 + 3 \cdot 1] = [2, 3]$.
- Multiply row 2 of A by each column of I: $[1 \cdot 1 + 4 \cdot 0, 1 \cdot 0 + 4 \cdot 1] = [1, 4]$.
- Multiplying any matrix by the identity matrix returns the original matrix.

4. Answer: $[[19,22],[43,50]]$

$$\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

- Row 1 $\times$ Col 1: $1 \cdot 5 + 2 \cdot 7 = 5 + 14 = 19$.
- Row 1 $\times$ Col 2: $1 \cdot 6 + 2 \cdot 8 = 6 + 16 = 22$.
- Row 2 $\times$ Col 1: $3 \cdot 5 + 4 \cdot 7 = 15 + 28 = 43$.
- Row 2 $\times$ Col 2: $3 \cdot 6 + 4 \cdot 8 = 18 + 32 = 50$.

5. Answer: $[[6,11],[14,6]]$

$$\begin{bmatrix} 6 & 11 \\ 14 & 6 \end{bmatrix}$$

- Product is 2×2 (outer dimensions: 2 rows, 2 cols).
- Entry (1,1): $1 \cdot 4 + 0 \cdot 2 + 2 \cdot 1 = 4 + 0 + 2 = 6$.
- Entry (1,2): $1 \cdot 1 + 0 \cdot 3 + 2 \cdot 5 = 1 + 0 + 10 = 11$.
- Entry (2,1): $3 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 12 + 2 + 0 = 14$.
- Entry (2,2): $3 \cdot 1 + 1 \cdot 3 + 0 \cdot 5 = 3 + 3 + 0 = 6$.

6. Answer: $[[11,3,2],[15,4,2]]$; order 2×3



$$\begin{bmatrix} 11 & 3 & 2 \\ 15 & 4 & 2 \end{bmatrix}$$

- Inner dimensions match (2=2), product order is 2×3 .
- Row 1: $[2 \cdot 4 + 1 \cdot 3, 2 \cdot 1 + 1 \cdot 1, 2 \cdot 0 + 1 \cdot 2] = [11, 3, 2]$.
- Row 2: $[3 \cdot 4 + 1 \cdot 3, 3 \cdot 1 + 1 \cdot 1, 3 \cdot 0 + 1 \cdot 2] = [15, 4, 2]$.

7. Answer: Not defined; B is 2×3 and A is 2×2 , inner dimensions $3 \neq 2$

- B is 2×3 and A is 2×2 .
- For $B \times A$, check inner dimensions: columns of B = 3, rows of A = 2.
- Since $3 \neq 2$, $B \times A$ is not defined.
- This also shows that matrix multiplication is NOT commutative in general.

8. Answer: $[[1,8],[0,9]]$

$$\begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}$$

- $A^2 = A \times A$.
- Entry (1,1): $1 \cdot 1 + 2 \cdot 0 = 1$.
- Entry (1,2): $1 \cdot 2 + 2 \cdot 3 = 2 + 6 = 8$.
- Entry (2,1): $0 \cdot 1 + 3 \cdot 0 = 0$.
- Entry (2,2): $0 \cdot 2 + 3 \cdot 3 = 0 + 9 = 9$.

9. Answer: $[[36],[12]]$

$$\begin{bmatrix} 36 \\ 12 \end{bmatrix}$$

- Step 1 – Compute $A \times B$: Row 1: $[1 \cdot 2 + 2 \cdot 1, 1 \cdot 1 + 2 \cdot 4] = [4, 9]$. Row 2: $[3 \cdot 2 + 0 \cdot 1, 3 \cdot 1 + 0 \cdot 4] = [6, 3]$. So $A \times B = [[4,9],[6,3]]$.
- Step 2 – Multiply $(A \times B) \times C$ (2×2 times 2×1): Entry (1,1): $4 \cdot 1 + 9 \cdot 2 = 4 + 18 = 22$. Wait, recalculate: Entry (1,1): $4 \cdot 1 + 9 \cdot 2 = 22$. Entry (2,1): $6 \cdot 1 + 3 \cdot 2 = 12$.
- Correct answer: $[[22],[12]]$.

10. Answer: $A \times B = [[16,11],[13,13]]$; $B \times A = [[11,19],[7,18]]$; $A \times B \neq B \times A$

- $A \times B$: (1,1)= $2 \cdot 5 + 3 \cdot 2 = 10 + 6 = 16$; (1,2)= $2 \cdot 1 + 3 \cdot 3 = 2 + 9 = 11$; (2,1)= $1 \cdot 5 + 4 \cdot 2 = 5 + 8 = 13$; (2,2)= $1 \cdot 1 + 4 \cdot 3 = 1 + 12 = 13$. So $A \times B = [[16,11],[13,13]]$.
- $B \times A$: (1,1)= $5 \cdot 2 + 1 \cdot 1 = 10 + 1 = 11$; (1,2)= $5 \cdot 3 + 1 \cdot 4 = 15 + 4 = 19$; (2,1)= $2 \cdot 2 + 3 \cdot 1 = 4 + 3 = 7$; (2,2)= $2 \cdot 3 + 3 \cdot 4 = 6 + 12 = 18$. So $B \times A = [[11,19],[7,18]]$.
- Since $[[16,11],[13,13]] \neq [[11,19],[7,18]]$, matrix multiplication is not commutative.

